

SIMATS ENGINEERING

ASSESSMENT TOOL 3 - LOW (Coding Challenge)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks: 50

ANNEXURE A

CODING CHALLENGE

Code of Conduct:

I _ B.Deepthi-(192424269) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Deepthi

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192424269RA

QUESTIONS

Q1
Array Index Traversal
5 marks

Q2
Null Pointer Handling
5 marks

Q3
String Comparison
5 marks

Q4
While Loop Correction
5 marks

Q5
Reverse String Logic
5 marks

Q6
Array Sum Calculation
5 marks

Q7
String Character Access
5 marks

Q1 Array Index Traversal5 marks

The following Java program is intended to print all elements of an array. However, it throws an error during execution. Identify and correct the mistake in the loop condition.

```
public class Q1_ArrayIndex {  
    public static void main(String[] args) {  
        int arr[] = {10, 20, 30};  
        for(int i = 0; i < arr.length; i++){  
            System.out.println(arr[i]);  
        }  
    }  
}
```

Your Code

```
1 import java.util.Scanner;  
2 public class Q1_ArrayIndex {  
3     public static void main(String[] args) {  
4         int arr[] = {10, 20, 30};  
5         for(int i = 0; i < arr.length; i++){  
6             System.out.println(arr[i]);  
7         }  
8     }  
9 }
```

Input

stdin input

Output

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Array Sum Calculation
5 marks

Q7
String Character Access
5 marks

Q2 Null Pointer Handling5 marks

A program attempts to print the length of a string but results in a runtime exception. Analyze the issue and modify the code to handle it properly.

```
public class Q2_NullPointer {  
    public static void main(String[] args) {  
        String str = null;  
        if(str != null)  
            System.out.println(str.length());  
        else  
            System.out.println("String is null");  
    }  
}
```

Your Code

```
1 public class Q2_NullPointer {  
2     public static void main(String[] args) {  
3         String str = null;  
4         if(str != null)  
5             System.out.println(str.length());  
6         else  
7             System.out.println("String is null");  
8     }  
9 }
```

Input

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String Character Access
5 marks

Q3 String Comparison5 marks

A developer compares two strings using the == operator, which leads to incorrect output. Identify the issue and correct the comparison method.

```
public class Q3_StringCompare {
    public static void main(String[] args) {
        String s1 = "Java";
        String s2 = new String("Java");
        if(s1.equals(s2)){
            System.out.println("Equal");
        }
    }
}
```

Your Code

```
1 import java.util.Scanner;
2 public class Q3_StringCompare {
3     public static void main(String[] args) {
4         Scanner s1 = new Scanner(System.in);
5         String s1 = s1.nextLine();
6         String s2 = new String("Java");
7         if(s1.equals(s2)){
8             System.out.println("Equal");
9         } else {
10            System.out.println("Not Equal");
11        }
12    }
13 }
```

Input

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Array Sum Calculation
5 marks

Q7
String Character Access
5 marks

Q4 While Loop Correction5 marks

The given program uses a loop to print numbers from 0 to 4, but it runs infinitely. Identify the cause and fix the loop.

```
public class Q4_Loop {
    public static void main(String[] args) {
        int i = 0;
        while(i < 5){
            System.out.println(i);
            i++;
        }
    }
}
```

Your Code

```
1 import java.util.Scanner;
2 public class Q4_Loop {
3     public static void main(String[] args) {
4         Scanner s1 = new Scanner(System.in);
5         int i = 0;
6         while (i < 5) {
7             System.out.println(i);
8             i++;
9         }
10    }
11 }
```

Input

stdin input

Output

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Coding Challenge

QUESTIONS

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Reverse String Logic
5 marks

Q6
Array Sum Calculation
5 marks

Q7

Q7 String Character Access

5 marks

A program tries to access a character from a string using an invalid index, causing an exception. Identify and correct the issue.

```
public class Q7_StringIndex {  
    public static void main(String[] args) {  
        String str = "Java";  
        System.out.println(str.charAt(3));  
    }  
}
```

Your Code

```
1 import java.util.Scanner;  
2 public class Q7_StringIndex {  
3     public static void main(String[] args) {  
4         String str = "Java";  
5         System.out.println(str.charAt(3));  
6     }  
7 }
```

Input

stdin input

Output

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Coding Challenge

QUESTIONS

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5 marks

Q5
Reverse String Logic
5 marks

Q6
Array Sum Calculation
5 marks

Q7
String Character Access

Q8

Q8 Case Insensitive Comparison

5 marks

A string comparison fails due to case sensitivity. Modify the program to make the comparison case-insensitive.

```
public class Q8_CaseInsensitive {  
    public static void main(String[] args) {  
        String str = "Java";  
        if(str.equalsIgnoreCase("JAVA")){  
            System.out.println("Match");  
        }  
    }  
}
```

Your Code

```
1 public class Q8_CaseInsensitive {  
2     public static void main(String[] args) {  
3         String str = "java";  
4  
5         if (str.equalsIgnoreCase("JAVA")) {  
6             System.out.println("Match");  
7         }  
8     }  
9 }
```

Input

stdin input

Output

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Coding Challenge

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QUESTIONS

Q1

Array Index Traversal

5 marks

✓

Q2

Null Pointer Handling

5 marks

✓

Q3

String Comparison

5 marks

✓

Q4

While Loop Correction

5 marks

✓

Q5

Reverse String Logic

5 marks

✓

Q6

Array Sum Calculation

5 marks

✓

Q7

String Character Access

5 marks

✓

Q9 Array Declaration

5 marks

The program contains an error in array declaration and initialization. Identify the syntax issue and correct it.

```
import java.util.Arrays;

public class Q9_ArrayDeclaration {
    public static void main(String[] args) {
        int arr[] = {1,2,3};
        System.out.println(Arrays.toString(arr));
    }
}
```

Your Code

```
1 import java.util.Arrays;
2
3 public class Q9_ArrayDeclaration {
4     public static void main(String[] args) {
5         int[] arr = {1, 2, 3};
6         System.out.println(Arrays.toString(arr));
7     }
8 }
```

Input

stdin input

Output

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QUESTIONS

Q1

Array Index Traversal

5 marks

✓

Q2

Null Pointer Handling

5 marks

✓

Q3

String Comparison

5 marks

✓

Q4

While Loop Correction

5 marks

✓

Q5

Reverse String Logic

5 marks

✓

Q6

Array Sum Calculation

5 marks

✓

Q7

String Character Access

5 marks

✓

Q8

✓

Q10 Switch Statement

5 marks

A switch statement prints multiple outputs due to missing statements. Identify the problem and correct the code.

```
public class Q10_Switch {
    public static void main(String[] args) {
        int num = 2;
        switch(num){
            case 1: System.out.println("One"); break;
            case 2: System.out.println("Two"); break;
            case 3: System.out.println("Three"); break;
        }
    }
}
```

Your Code

```
1 public class Q10_Switch {
2     public static void main(String[] args) {
3         int num = 2;
4         switch(num){
5             case 1: System.out.println("One"); break;
6             case 2: System.out.println("Two"); break;
7             case 3: System.out.println("Three"); break;
8         }
9     }
10 }
```

Input

stdin input

Output